

CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS
Higher 1

CANDIDATE
NAME

CLASS

CHEMISTRY

Paper 2

8872/02

Tuesday 21 August 2012
2 hours

Candidates answer Section A on the Question Paper

Additional Materials: Answer Paper

Data Booklet

Graph Paper (2 sheets)

READ THESE INSTRUCTIONS FIRST

Write your name and HT group on all the work you hand in.

Write in dark blue or black pen in the spaces provided.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** the questions

Section B

Answer **two** questions on separate answer paper.

You are advised to spend not more than 1 hour for Section B.

You are advised to show all working in calculations.

You may use a calculator.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Paper 1	30
Section A	40
A1	/16
A2	/14
A3	/10
Section B	40
B4	/20
B5	/20
B6	/20
TOTAL	110

This document consists of **12** printed pages and **0** blank page.

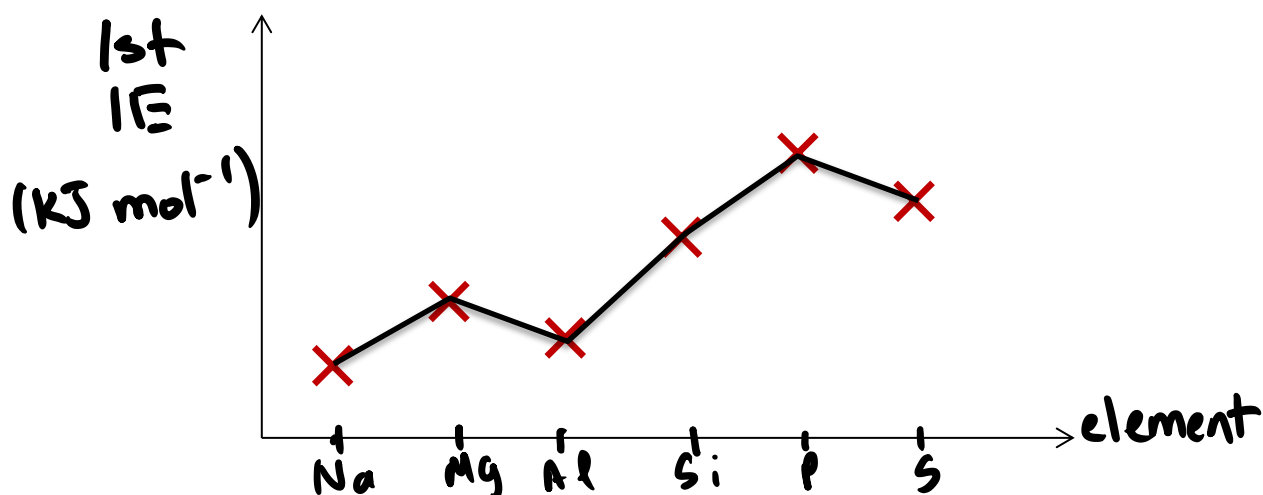
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Section A

Answer **all** questions in this section in the spaces provided.

- 1 Elements of the Periodic Table have certain trends in their physical and chemical properties.

(a) Sketch a graph showing the variation of first ionisation energy of elements from sodium to sulfur, and explain the general trend qualitatively.



First ionisation energy generally increases from Na to S due to the increase in nuclear charge and decrease in atomic size, while the number of electron shell remains the same, hence shielding effect remains almost the same.

[3]

- (b) (i) Arrange magnesium, silicon and phosphorus in the order of increasing melting point.

Phosphorus, Magnesium, Silicon

- (ii) Explain in terms of structure and bonding your answer in (b)(i)

Si has the highest melting point as it has a giant covalent structure, and strong covalent bonds between atoms.

Mg has metallic bonds which are electrostatic attraction between the sea of electrons and cations, while P exists as simple covalent molecules with van der Waals' forces.

Thus more energy is required to overcome the covalent bonds between Si atom than the metallic bond of Mg, while the least energy is required for the van der waal's forces of attraction between P molecules.

[4]

- (c) (i) Describe the reactions of sodium and sulphur respectively as they are burnt in a stream of oxygen. For each reaction, write a balanced equation and state what would be observed.

Sodium, Equation: $4\text{Na} + \text{O}_2 \rightarrow 2\text{Na}_2\text{O}$

Observations: Sodium burns in yellow flame to give a white solid.

Sulfur, Equation: $\text{S} + \text{O}_2 \rightarrow \text{SO}_2$

Observations: Sulfur burns in pale blue flame to give a colourless gas.

- (ii) State the colour that you would see when each of the **products obtained** in (c)(i) is separately added to water containing Universal Indicator.

Sodium: Blue Sulfur: Red or Orange [6]

- (d) The molecules of nitrogen oxides are very unique in their bondings and often involve dative bonds.

- (i) Draw a dot and cross diagram of the following nitrogen oxides

I. N_2O

II. NO_2

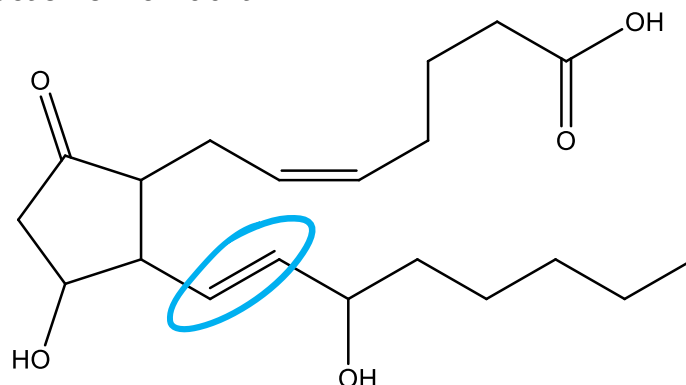


- (ii) Suggest the bond angle for NO_2 : 120° OR more than 120° but less than 180°

[3]

[Total: 16]

- 2 Organic compounds are often synthesised and used in modern medicine, with many of these compounds first discovered from natural sources or origins. For example, naturally occurring prostaglandins was discovered in 1920's and their chemical structures deduced in the 1950's. For the research of its biochemical processes and functions, the scientists were awarded the 1982 Nobel Prize for Medicine. One form of prostaglandins, which is marketed as Prostin E2, is now used for induction of? childbirth.



Prostin E2

- (a) Alkenes can exist as either cis or trans isomers. In the diagram above, circle the alkene.
- (b) Name two functional groups, other than alkene and carboxylic acid, which are present in Prostin E2.
- 1 Secondary alcohol
 - 2 Ketone
- [3]
- (c) In an attempt to deduce the chemical structure of **Prostin E2**, chemical tests can be used to identify individual functional groups. But the test used must **only** react with the desired functional group and **not** the others. Suggest the reagents, conditions and observations for identification of:

I. Alkene

Reagents and conditions: Aqueous bromine / bromine liquid at room temperature

Observations: Orange brown aqueous bromine is decolourised

.....

II. Another functional group that you have named in (b)

Functional Group: Secondary Alcohol / Ketone

Reagents and conditions: $K_2Cr_2O_7$ with dil. H_2SO_4 heat / 2,4-dinitrophenylhydrazine warm

Observations: Orange $K_2Cr_2O_7$ turns green in colour / Orange precipitate observed.

..... [4]

- (d) Prostin E2 has 20 carbons which can be extended to 21 carbons when it undergoes a suitable reaction.

(i) State the reagent and condition for this reaction.

..... HCN with trace amount of NaOH at 10 – 20 °C

(ii) Suggest the type of reaction.

..... Addition [2]

- (e) In order to allow drugs to be separated from other organic side-products during synthesis, the drugs can be made more soluble in aqueous solution with suitable reagents. Prostin E2 has a carboxylic acid functional group, which can undergo further reaction for this purpose.

(i) Suggest **two** reagents that can **only** react with the carboxylic acid group to result in a more soluble compound and construct balanced equations for these reactions, using RCOOH to represent Prostin E2.

1 $RCOOH + NaOH \rightarrow RCOO^- Na^+ + H_2O$

2 $2RCOOH + Na_2CO_3 \rightarrow 2RCOO^- Na^+ + CO_2 + H_2O$

(ii) Suggest why the products in (e)(i) are more soluble than Prostin E2.

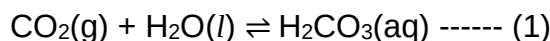
..... The carboxylate ion formed can form ion-dipole interaction with water which is stronger / more favourable than the hydrogen bonding of carboxylic acid with

(iii) Suggest which of the 2 reagents you have suggested in (e)(ii) is **not** preferred in industrial process and why is that so.

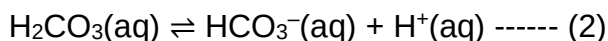
..... Na_2CO_3 is less preferred as CO_2 is released which can result in elevated greenhouse effect. [4]

[Total: 14 marks]

- 3 Carbonic acid, H_2CO_3 , is involved in the transport of carbon dioxide, CO_2 , out of the body via respiratory gas exchange. Carbon dioxide is quickly dissolved in the blood plasma to form carbonic acid, H_2CO_3 , in the presence of carbonic anhydrase, an enzyme found in red blood cells.



The resultant carbonic acid formed then dissociates into hydrogencarbonate ion, HCO_3^- , and hydrogen ion, H^+ , in the following equilibrium.



This reaction is reversed in the lungs, where hydrogencarbonate is converted back into carbon dioxide and allowed to expel. This hydrogencarbonate/carbonic acid buffer helps to maintain our blood at pH 7.4.

The Henderson–Hasselbalch equation can be used to calculate the pH of blood using the constituents of the hydrogencarbonate buffer, namely concentration of bicarbonate ions and carbonic acid. The equation is shown below.

$$\text{pH} = \text{pK}_a(\text{H}_2\text{CO}_3) + \lg \frac{[\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]}$$

A modified Henderson–Hasselbalch equation, shown below, can also be used to calculate the pH of blood.

$$\text{pH} = \text{pK}_a(\text{H}_2\text{CO}_3) + \lg \frac{[\text{HCO}_3^-]}{[0.03 \times \text{p}_{\text{CO}_2}]}$$

The calculation is based on concentration of hydrogencarbonate (in mol dm^{-3}) and pressure of carbon dioxide (in atm), which is represented as p_{CO_2} . Carbonic acid has an acid dissociation constant, K_a , of $4.27 \times 10^{-7} \text{ mol dm}^{-3}$.

During strenuous exercise, lactic acidosis, a condition whereby pH is low in body tissues and blood, may occur. Cells receive too little oxygen and the lowered respiration rates of the cells cause the production of lactic acid to provide energy required by body. This leads to a lower pH levels in blood. To increase blood pH, the body breathes more to reduce the acidity.

In some cases, breathing is not sufficient to reduce the acidity of the blood and pH will continue to fall. If the pH of blood falls under 7.1, it will be considered an emergency and may warrant treatment with intravenous bicarbonate solution. Bicarbonate is given at 0.050 - 0.100 mol at a time under close monitoring of the arterial blood gas readings.

- (a) State the purpose of carbonic anhydrase in the red blood cell. [1]

Biological catalyst

- (b) Using relevant equations, describe how the hydrogencarbonate/carbonic acid buffer resist small pH changes in the blood. [2]

If the blood becomes too acidic, $\text{HCO}_3^- + \text{H}^+ \rightarrow \text{H}_2\text{O} + \text{CO}_2$

If the blood becomes too alkaline, $\text{H}_2\text{CO}_3 + \text{OH}^- \rightarrow \text{HCO}_3^- + \text{H}_2\text{O}$

Thus, pH of blood remained relatively constant.

- (c) With reference to the equilibrium 1 and 2, explain how breathing more reduces the acidity of blood. [3]

When one breathes more, more $\text{CO}_2(\text{g})$ is released and position of equilibrium

1 will shift to the left. Thus reducing the concentration of $\text{H}_2\text{CO}_3(\text{aq})$

When concentration of $\text{H}_2\text{CO}_3(\text{aq})$ is reduced, it will cause position of equilibrium 2 to shift to the left.

Thus, H^+ is removed from the blood and acidity of the blood is decreased.

- (d) Given that $2.33 \times 10^{-3} \text{ mol dm}^{-3}$ of carbonic acid is present in blood at equilibrium, calculate the concentration of bicarbonate ions present. [2]

$$\begin{aligned} \text{pH} &= \text{pK}_a(\text{H}_2\text{CO}_3) + \lg \frac{[\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]} \\ 7.4 &= -\lg(4.27 \times 10^{-7}) + \lg \frac{[\text{HCO}_3^-]}{2.33 \times 10^{-3}} \\ [\text{HCO}_3^-] &= 2.50 \times 10^{-2} \text{ mol dm}^{-3} \end{aligned}$$

- (e) A student carried out strenuous exercises immediately from rest. Given that 0.0939 atm of carbon dioxide is present and using your answer in (d), calculate the pH of his blood. Hence, conclude if an emergency has occurred. [2]

$$\begin{aligned} \text{pH} &= \text{pK}_a(\text{H}_2\text{CO}_3) + \lg \frac{[\text{HCO}_3^-]}{[0.03 \times p_{\text{CO}_2}]} \\ &= -\lg(4.27 \times 10^{-7}) + \lg \frac{[2.5 \times 10^{-2}]}{[0.03 \times 0.0939]} \\ &= 7.32 \text{ (3.s.f.)} \end{aligned}$$

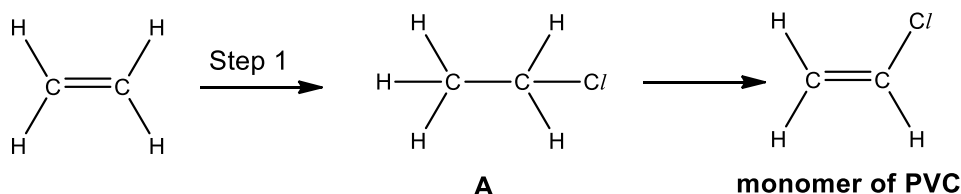
An emergency has not occurred.

Section B

Answer **two** questions from this section on separate answer sheet.

- 4 (a)** Halogenoalkanes have a variety of commercial uses. For example, PVC is a polymer commonly used in insulation of electrical wiring.

To produce the monomer of PVC, the following reaction steps are carried out industrially starting with ethene.



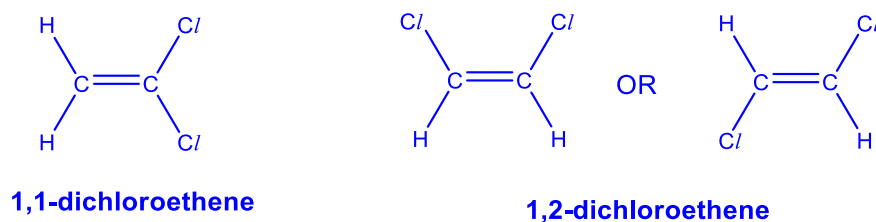
- (i)** State the reagent used in Step 1 and state the type of reaction that has occurred. .

HCl(g), Addition Reaction

- (ii)** Suggest the reagent and conditions needed if monomer of PVC is to be formed directly (in a single step) from ethene.

Cl₂(g) and UV light

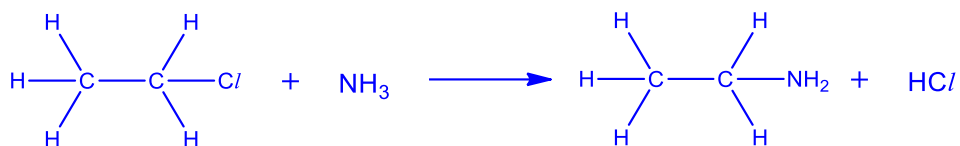
- (iii)** The reaction in **part (ii)** may result in 2 structural isomers with the molecular formula C₂H₂Cl₂. Draw both isomers and label with the IUPAC names.



- (iv)** Ethene can also be converted to ethanol, state the reagent and conditions for this conversion in a school laboratory. [6]

*1) Cold **Concentrated** H₂SO₄ followed by 2) H₂O, **warm**.*

- (b)** Compound A in part (a) can also undergo a variety of reactions. It can react with ammonia under certain conditions. Write an equation for this and state the necessary conditions. [2]



NH₃ in **ethanol/alcohol** and **heated in sealed tube**.

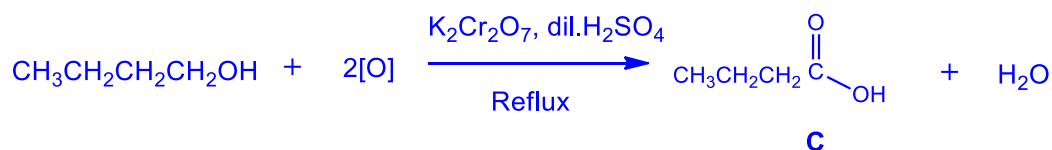
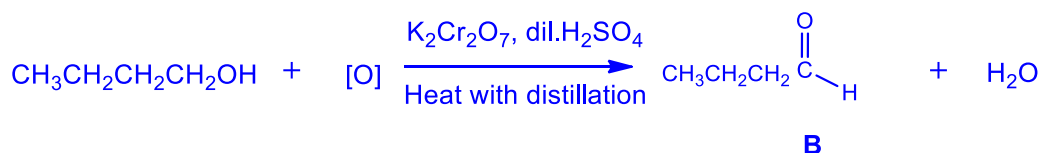
- (c) Halogenoalkanes can also react to give alcohols. CH₃CH₂CH₂CH₂Br and CH₃CH₂CH₂CH₂Cl can both give CH₃CH₂CH₂CH₂OH. With reference to the Data Booklet, suggest the relative rates of each reaction. [2]

C-Cl is a **stronger bond** with bond energy of **340 kJ mol⁻¹** and while C-Br is **weaker** with bond energy of C-Br **280 kJ mol⁻¹**, hence **easier to break**.

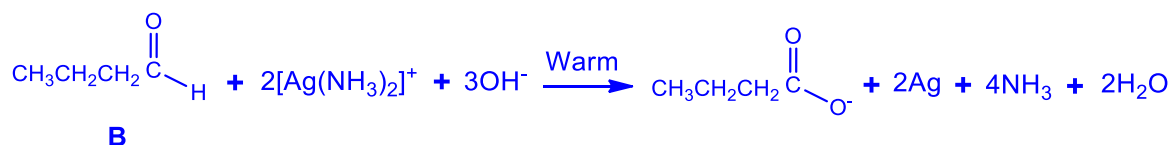
Hence the rate of reaction for CH₃CH₂CH₂CH₂Cl will be **slower** than that of CH₃CH₂CH₂CH₂Br.

- (d) CH₃CH₂CH₂CH₂OH can be oxidised to two different products, **B** and **C**. **B** reacts with Tollens' Reagent, while **C** reacts with sodium carbonate.

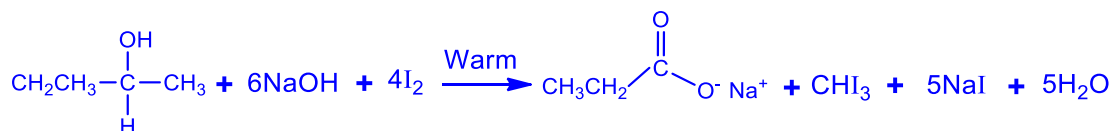
- (i) Identify **B** and **C**, and write balanced equations for formations of **B** and **C** with the suitable reagents and conditions for each case.



- (ii) Write a balanced equation for the reaction of **B** with Tollens' Reagent. [4]



- (e) An isomer of CH₃CH₂CH₂CH₂OH can react with aqueous iodine in sodium hydroxide to give a yellow precipitate. Suggest the structure of this isomer and write a balanced equation for this reaction. [2]



- (f) Halogenated carboxylic acid has different properties as compared to carboxylic acid. Using $\text{CH}_3\text{CHClCO}_2\text{H}$ and $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$, suggest one physical and one chemical property that is different, and briefly explain why is that so. [4]

One physical property is that $\text{CH}_3\text{CHClCO}_2\text{H}$ has higher boiling point as compared to $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$. This is because Cl has more electrons than H, thus $\text{CH}_3\text{CHClCO}_2\text{H}$ has a larger electron cloud, which results in more extensive Van der Waal's forces of attraction, and higher boiling point.

One chemical property is that $\text{CH}_3\text{CHClCO}_2\text{H}$ is a stronger acid as compared to $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$.

This is because the Cl is electron withdrawing, thus the O-H bond in $\text{CH}_3\text{CHClCO}_2\text{H}$ is more polar and weaker and dissociates to larger extent to give H^+ .

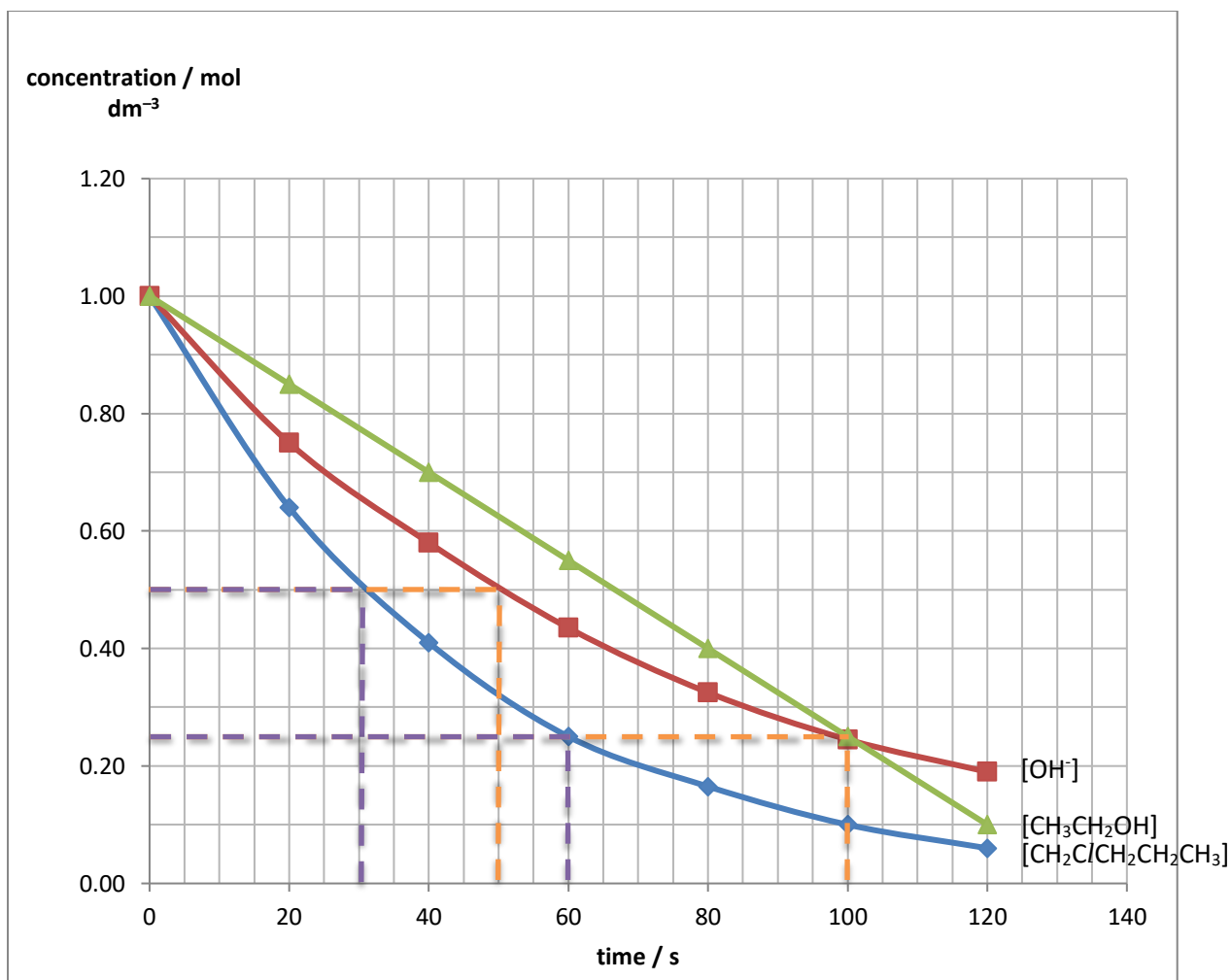
- 5 (a)** An elimination reaction is a type of organic reaction in which two substituents are removed from a molecule. An important class of elimination reactions are those involving alkyl halides, in the presence of base and alcohol to form alkenes.

To investigate how the rate of reaction depends on the concentration of each of the three reactants, three experiments were carried out. In each experiment, the concentrations of two reactants were in excess and kept constant, whilst the concentration of the third reactant was measured over time. For example, in experiment 1, $[\text{OH}^-]$ and $[\text{CH}_3\text{CH}_2\text{OH}]$ are in large excess, while $[\text{CH}_2\text{Cl}/\text{CH}_2\text{CH}_2\text{CH}_3]$ is measured.

Data from the experiments are shown in the table below.

	Experiment 1	Experiment 2	Experiment 3
Time / s	$[\text{CH}_2\text{Cl}/\text{CH}_2\text{CH}_2\text{CH}_3] / \text{mol dm}^{-3}$	$[\text{OH}^-] / \text{mol dm}^{-3}$	$[\text{CH}_3\text{CH}_2\text{OH}] / \text{mol dm}^{-3}$
0	1.000	1.000	1.000
20	0.640	0.750	0.850
40	0.410	0.580	0.700
60	0.250	0.435	0.550
80	0.165	0.325	0.400
100	0.100	0.245	0.250
120	0.060	0.190	0.100

- (i)** Using the data given, plot three graphs on the same axes and label your graphs clearly.



- (ii) Using your graph, derive the order of reaction for each reactant. Explain your answers briefly.

Order of reaction of ethanol = 0

as the rate (gradient) of the reaction is constant as the reaction proceeds.

Order of reaction of base = 1

There is a constant half-life of 50s

Order of reaction of 1-chlorobutane = 1

There is a constant half-life of 30s

- (iii) Use your answer from (ii), write a rate equation for the reaction. State the units for the rate constant. [12]

$$\text{Rate} = k[\text{CH}_2\text{ClCH}_2\text{CH}_2\text{CH}_3][\text{OH}^-]$$

$$\text{Units of } k: \text{mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$$

(b) Upon elimination of 1-chlorobutane, hydrogen chloride is produced as a by-product. It will then react with sodium hydroxide present in the solution.

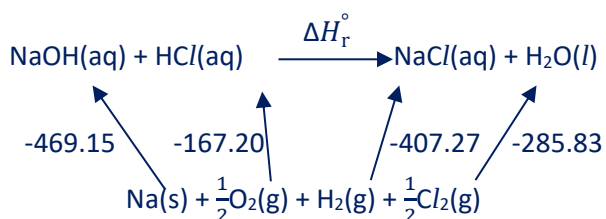
(i) State the type of reaction that has occurred and write an equation.

Acid – base neutralisation / reaction



(ii) Hence, calculate the enthalpy change of reaction in (b)(i) given the following enthalpies:

Compound	$\Delta H_f^\circ / \text{kJ mol}^{-1}$
NaOH(aq)	-469.15
NaCl(aq)	-407.27
HCl(aq)	-167.20
H ₂ O(l)	-285.83



By Hess' Law,

$$(-469.1) + (-167.2) + \Delta H_r^\circ = (-407.27) + (-285.83)$$

$$\begin{aligned}
 \Delta H_r^\circ &= -693.1 - (-636.3) \\
 &= -56.8 \text{ kJ mol}^{-1}
 \end{aligned}$$

(iii) Using your answer in (b)(ii), state the enthalpy change of neutralisation, ΔH_n° , of the reaction when barium hydroxide, Ba(OH)₂, which acts as a strong base, is reacted with hydrogen chloride instead. [6]

$$\Delta H_n^\circ = -56.8 \text{ kJ mol}^{-1}$$

- (c) To obtain concentrated sodium hydroxide solution, sodium hydroxide pellets are dissolved in water. The enthalpy change of dissolving the pellets can be calculated by measuring the temperature change in the solution. This enthalpy change is known as enthalpy change of solution, ΔH_{sol}° .

In an experiment, 1.0 g of sodium hydroxide pellets were dissolved in 50 cm³ of water. It was found that the temperature rose by 5.32 K. Given that the specific heat capacity of water is 4.18 kJ kg⁻¹ K⁻¹, calculate the enthalpy change of solution of sodium hydroxide. [2]

$$\text{Heat change, } Q = mc\Delta T = 50 \times 4.18 \times (-5.32) = -1.112 \text{ kJ}$$

$$\text{No. of mole NaOH} = \frac{1}{23+16+1} = 0.0250 \text{ mol}$$

$$\Delta H_{sol}^{\circ} = \frac{-1.112}{0.025} = -44.5 \text{ kJ mol}^{-1}$$

6 Phosphorus containing compounds are important industrial chemical as they are used in the manufacturing of organophosphorus compounds which have a wide variety of applications in herbicides, insecticides, oil additives, and flame retardants.

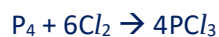
(a) A certain phosphorus containing compound is synthesised commercially by refluxing white phosphorus, P_4 , with chlorine gas, Cl_2 . It was found that 2 g of phosphorus reacts with exactly 2.32 dm^3 of Cl_2 (measured at r.t.p.) and no other products are formed in the reaction.

(i) Derive the stoichiometric ratio of phosphorus to chlorine and show that the compound synthesised is phosphorus trichloride, PCl_3 , by writing a balanced equation for the reaction

$$\text{No of mole of } P_4 = \frac{2}{31 \times 4} = 0.0161 \text{ mol}$$

$$\text{No. of mole of } Cl_2 = \frac{2.32}{24} = 0.0967 \text{ mol}$$

$$\text{Ratio of } P_4 : Cl_2 = 1:6$$



(ii) Phosphorus trichloride is also used to produce a phosphorus containing acid by hydrolysis in water. Write an equation for this reaction.



(iii) Using the equations from (a)(i) and (ii), calculate mass of P_4 required to produce a 1 dm^3 of $0.250 \text{ mol dm}^{-3}$ solution of the acid in part (ii). [6]

$$\text{From both equations: } \frac{1}{4}P_4 \equiv PCl_3 \equiv H_3PO_3$$

In 1 dm^3 , $\frac{0.250}{4} \text{ mol}$ of P_4 is required to produce a $0.250 \text{ mol dm}^{-3}$ H_3PO_3 solution

$$\text{Mass of } P_4 \text{ required} = 0.0625 \times (31 \times 4) = 7.75 \text{ g}$$

(b) Draw the shape of phosphorus trichloride. [1]



- (c) (i) Similar to phosphorus trichloride, aluminium chloride, $AlCl_3$, undergoes hydrolysis to give an acidic solution. Show, by means of two equations, how aluminium chloride reacts when added to water.



- (iii) At the boiling point of aluminium chloride, its molar mass was found to be 267 g mol^{-1} . By considering the structure and bonding, explain why the boiling point of aluminium chloride is higher than that of phosphorus trichloride.

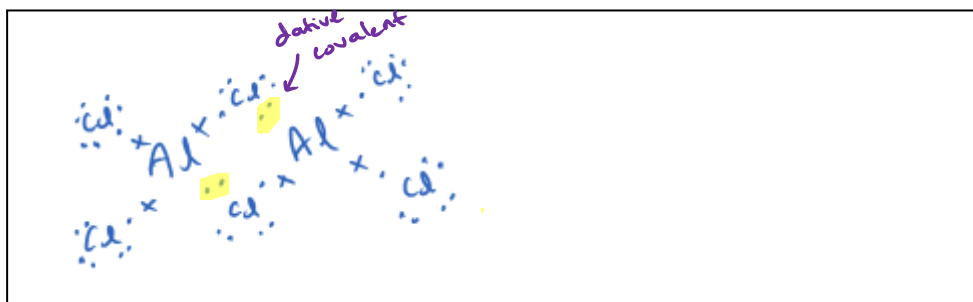
$AlCl_3$ and PCl_3 are both **simple covalent compound**.

PCl_3 has **permanent dipole – permanent dipole interaction** between its molecules while $AlCl_3$ has **weak van der waal's forces of attraction**.

However, molar mass of $AlCl_3 = 133.5 \text{ g mol}^{-1}$, thus it **dimerises** at elevated temperature as the molar mass is twice the value.

Thus, van der waal's forces of attraction is **more extensive** due to the larger electron cloud. This results in the higher boiling point of $AlCl_3$ than PCl_3 .

- (iii) Draw the dot-and-cross diagram of aluminium chloride at its boiling point. [6]



- (e) Aluminium is in Group III and phosphorus is in Group V of the periodic table, but both of these elements are in Period three.

- (i) Write the electronic configurations of their most stable ions Al^{3+} and P^{3-} .



- (ii) Explain the differences in the first ionisation energy between aluminium and phosphorus and the ionic radius of their respective ions. [7]

[Total: 20 marks]

Ionic radius:

Ionic radius of Al^{3+} is **smaller** than that of P^{3-} . Al^{3+} has 2 shells of electrons while P^{3-} has 3 shells.

Thus Al^{3+} has a **smaller shielding effect** than P^{3-} . This is **more significant than the smaller nuclear charge** of Al^{3+}

, thus it has a **stronger attraction for its valence electrons**, resulting in a smaller ionic radius.

First ionisation energy:

1st IE of P is **greater** than that of Al.

Al: $1s^2 2s^2 2p^6 3s^2 3p^1$

P: $1s^2 2s^2 2p^6 3s^2 3p^3$

Electrons in P experience a **greater nuclear charge** than Al as P has a greater number of protons in the nucleus.

The shielding effect is similar for both Al and P.

Thus **more** energy is required to remove an electron in P than in Al.